Typediff

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Typediff

A browser-based tool enabling rapid exploration of all types from the type hierarchy involved in the parsing of input text.

The Aim

- Identify types involved in the analysis of specific syntactic phenomena
- Intuition is to contrast phenomenon positive sentences with similar phenomenon negative sentences, ie "minimal pairs"

Applications

- Phenomenon detection
- Phenomenon-based navigation of TDL files
- Grammar documentation
- Others?

Methodology

1. Input sentences/fragments parsed with ACE
2. Every type from full AVM of the best parse is extracted
3. Each input fragment is converted into a set of type names
4. Positive (A) fragments contrasted with negative (B) fragments using set difference
5. Resultant types organized by HPSG theoretic categories (eg `sign`, `synsem`, `head` etc) and sorted by % coverage over treebank

Example input for exploring verb phrase ellipsis:

A *Kim likes fish, but I don't.* +

B *Kim likes fish, but I don't like fish.* —

Other features

- Additional inputs if phenomenon not amenable to minimal pair
- Relevant tree(s) can be selected from the parse forest
- Can show differences across inherited supertypes
- Supports diffing trees
  - from within parse forest of sentence
  - from across different grammars
- Interfaces with existing DELPH-IN resources
  - Type coverage taken from treebanks
  - Lexical Type Database
  - Fangorn (Treebank Search)

Grammars supported

- ERG, GG, Jacy, HaG, NorSource, INDRA
- Requirements: Grammar configured for ACE

Homepage: `http://moin.delph-in.net/TypediffTop`

Live demo: `http://hum.csse.unimelb.edu.au/typediff`